Q1

```
mysql> use cdac;
Database changed
mysql> delimiter //
mysql> create procedure check_even_odd
    -> (IN n int)
    -> BEGIN
    -> IF n%2 = 0 THEN
    -> set @status = 'EVEN';
    -> Else
    -> SET @status = 'ODD';
    -> END IF;
    -> select @status as res;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql> DELIMITER ;
mysql> call check_even_odd(25);
+------+
| res  |
+------+
| ODD  |
+------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> call check_even_odd(40);
+------+
| res  |
+------+
| EVEN |
+------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

Q2

```
mysql> delimiter //
mysql> create procedure calculate_interest (IN p int, IN rate int, IN time int)
    -> BEGIN
    -> declare res decimal(7,2)
    -> ;
    -> set res = (p * rate * time)/100;
    -> select res as intrest;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql> Delimiter ;
mysql> call calculate_interest(10000, 5, 2);
+---------+


| intrest |
+---------+
| 1000.00 |
+---------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> call calculate_interest(25000, 7, 3);
+---------+
| intrest |
+---------+
| 5250.00 |
+---------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

Q3

```
mysql> delimiter //
mysql> create procedure calculate_grade(IN rollno int, IN name varchar(20), IN marks int)
    -> BEGIN
    -> if marks > 90 then
    -> set @res = 'A+';
    -> elseif marks < 89 AND marks > 80 then
    -> set @res = 'A';
    -> elseif marks < 79 AND marks > 70 then
    -> set @res = 'B';
```

```
    -> elseif marks < 69 AND marks > 60 then
    -> set @res = 'C';
    -> elseif marks < 59 AND marks > 50 then
    -> set @res =  'D';
    -> else
    -> set @res = 'F';
    -> end if;
    -> select rollno, name, marks, @res as result;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> CALL calculate_grade(101, 'Amit', 85);
    -> //
+--------+------+-------+--------+
| rollno | name | marks | result |
+--------+------+-------+--------+
|    101 | Amit |    85 | A      |
+--------+------+-------+--------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_grade(102, 'Priya', 72)//
+--------+-------+-------+--------+
| rollno | name  | marks | result |
+--------+-------+-------+--------+
|    102 | Priya |    72 | B      |
+--------+-------+-------+--------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_grade(103, 'Rahul', 45)//
+--------+-------+-------+--------+
| rollno | name  | marks | result |
+--------+-------+-------+--------+
|    103 | Rahul |    45 | F      |
+--------+-------+-------+--------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

Q4

```
mysql> create procedure calculate_bonus(IN emp_name varchar(20), IN emp_salary int)
    -> BEGIN
    -> if emp_salary >= 100000 then
    -> set @res = 0.20;
    -> elseif emp_salary >= 50000 then
    -> set @res = 0.15;
    -> elseif emp_salary >= 30000 then
    -> set @res = 0.10;
    -> else
    -> set @res = 0.05;
    -> end if;
    -> select emp_name as employee, @res * 100 as bonus_percentage;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql> DELIMITER ;
mysql> call calculate_bonus('Ryan', 120000);
+----------+-----------------------------------+
| employee | bonus_percentage                  |
+----------+-----------------------------------+
| Ryan     | 20.000000000000000000000000000000 |
+----------+-----------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> call calculate_bonus('Paige', 65000);
+----------+-----------------------------------+
| employee | bonus_percentage                  |
+----------+-----------------------------------+
| Paige    | 15.000000000000000000000000000000 |
+----------+-----------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> call calculate_bonus('Rose', 25000);
+----------+-----------------------------------+
| employee | bonus_percentage                  |
+----------+-----------------------------------+
| Rose     | 5.000000000000000000000000000000 |
+----------+-----------------------------------+
1 row in set (0.00 sec)
```

Q5

```
mysql> create procedure find_maximum(IN a int, IN b int, IN c int)
    -> BEGIN
    -> if a > b AND a > c then
    -> set @maximum = a;
    -> if b > a AND b > c then
    -> set @maximum = b;
    -> elseif c > a AND c > b then
    -> set @maximum = c;
    -> end^C
mysql> create procedure find_maximum(IN a int, IN b int, IN c int)
    -> BEGIN
    -> if a > b AND a > c then
    -> set @maximum = a;
    -> elseif b > a AND b > c then
    -> set @maximum = b;
    -> elseif c > a AND c > b then
    -> set @maximum = c;
    -> end if;
    -> select @maximum as max_value;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> CALL find_maximum(25, 78, 45)//
+-----------+
| max_value |
+-----------+
|        78 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL find_maximum(100, 50, 75);//
+-----------+
| max_value |
+-----------+
|       100 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

```
mysql> CALL find_maximum(30, 30, 20);//
+-----------+
| max_value |
+-----------+
|       100 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> delimiter ;
mysql> CALL find_maximum(100, 50, 75);
+-----------+
| max_value |
+-----------+
|       100 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL find_maximum(30, 30, 20);//
+-----------+
| max_value |
+-----------+
|       100 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

    -> ^C
mysql> CALL find_maximum(30, 30, 20);
+-----------+
| max_value |
+-----------+
|       100 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> call find_maximum(40, 20, 10);
+-----------+
```

```
| max_value |
+-----------+
|        40 |
+-----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql>
```

Q6.

```
mysql> create procedure salary_analysis(
    -> IN emp_name varchar(30),
    -> IN emp_salary int
    -> )
    -> BEGIN
    -> IF emp_salary >= 100000 then
    -> set @cat = 'High';
    -> set @allowance = 0.20;
    -> elseif emp_salary >= 50000 then
    -> set @cat = 'Medium';
    -> set @allowance = 0.15;
    -> else
    -> set @cat = 'Low';
    -> set @allowance = 0.10;
    -> end if;
    -> select emp_name, emp_salary, @cat as category, @allowance * 100 as
allowance_percentage, (emp_salary * @allowance) as allowance_amount, emp_salary +
@allowance as final_salary;
    -> END //
Query OK, 0 rows affected (0.06 sec)

mysql> delimiter ;
mysql> call salary_analysis('Ana', 120000);
+----------+------------+----------+---------------------------------+-----------------------------------+------------
--------------------------+
| emp_name | emp_salary | category | allowance_percentage            | allowance_amount
| final_salary                      |
+----------+------------+----------+---------------------------------+-----------------------------------+------------
--------------------------+
| Ana      |     120000 | High     | 20.0000000000000000000000000000 |
24000.0000000000000000000000000000 | 120000.2000000000000000000000000000 |
+----------+------------+----------+---------------------------------+-----------------------------------+------------
--------------------------+
1 row in set (0.04 sec)

Query OK, 0 rows affected (0.05 sec)
       call salary_analysis('Ana', 120000);^C
mysql> call salary_analysis('Nancy', 65000);
+----------+------------+----------+---------------------------------+-----------------------------------+------------
-------------------------+
| emp_name | emp_salary | category | allowance_percentage            | allowance_amount
| final_salary                      |
```

```
+----------+------------+----------+---------------------------------+-----------------------------------+-------------
------------------------+
| Nancy    |      65000 | Medium   | 15.00000000000000000000000000000000 |
9750.00000000000000000000000000000000 | 65000.15000000000000000000000000000000 |
+----------+------------+----------+---------------------------------+-----------------------------------+-------------
------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> call salary_analysis('yami', 35000);
+----------+------------+----------+---------------------------------+-----------------------------------+-------------
------------------------+
| emp_name | emp_salary | category | allowance_percentage            | allowance_amount
| final_salary                    |
+----------+------------+----------+---------------------------------+-----------------------------------+-------------
------------------------+
| yami     |      35000 | Low      | 10.00000000000000000000000000000000 |
3500.00000000000000000000000000000000 | 35000.10000000000000000000000000000000 |
+----------+------------+----------+---------------------------------+-----------------------------------+-------------
------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql>
```

Q7

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE calculate_electricity_bill(
    ->     IN consumer_name VARCHAR(30),
    ->     IN consumer_units INT
    -> )
    -> BEGIN
    ->     DECLARE rate INT;
    ->
    ->     IF consumer_units > 400 THEN
    ->         SET rate = 7;
    ->     ELSEIF consumer_units > 200 THEN
    ->         SET rate = 5;
    ->     ELSEIF consumer_units > 100 THEN
    ->         SET rate = 3;
    ->     ELSE
    ->         SET rate = 2;
    ->     END IF;
    ->
    ->     SELECT consumer_name AS Consumer_Name,
    ->            consumer_units AS Units_Consumed,
    ->            rate * consumer_units AS Total_Bill_Amount;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> call calculate_electricity_bill('Amy', 80);
+---------------+---------------+-------------------+
| Consumer_Name | Units_Consumed | Total_Bill_Amount |
+---------------+---------------+-------------------+
| Amy           |             80 |               160 |
+---------------+---------------+-------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> call calculate_electricity_bill('peter', 150);
+---------------+---------------+-------------------+
| Consumer_Name | Units_Consumed | Total_Bill_Amount |
+---------------+---------------+-------------------+
| peter         |            150 |               450 |
+---------------+---------------+-------------------+
```

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```
mysql> call calculate_electricity_bill('Ross', 350);
+---------------+----------------+-------------------+
| Consumer_Name | Units_Consumed | Total_Bill_Amount |
+---------------+----------------+-------------------+
| Ross          |            350 |              1750 |
+---------------+----------------+-------------------+
```

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```
mysql> call calculate_electricity_bill('Sana', 500);
+---------------+----------------+-------------------+
| Consumer_Name | Units_Consumed | Total_Bill_Amount |
+---------------+----------------+-------------------+
| Sana          |            500 |              3500 |
+---------------+----------------+-------------------+
```

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)